Method	Pretraining	Classification (macro-F1)	Detection (mAP)
Chance level	–	10.00	7.84
Classical acoustic features			
Spectral features	–	14.15 ± 0.67	26.34 ± 1.17
MFCCs	–	21.22 ± 0.99	33.91 ± 1.77
Mean spectrogram	–	32.05 ± 1.30	47.80 ± 2.10
Off-the-shelf pretrained models			
AVES-core	General audio (AudioSet, FSD50K)	51.60 ± 2.28	57.39 ± 2.07
BioLingual	Audio-text (AnimalSpeak)	54.03 ± 2.72	66.52 ± 2.23
AVES-bio	Animal vocalizations (AudioSet, VGGSound)	54.45 ± 2.58	65.01 ± 2.29
Pretrained on OpenWhistle (ours)			
AVES-bio‡	OpenWhistle (from scratch)	60.97 ± 3.76	76.17 ± 1.97
Wav2Vec2.0	OpenWhistle (from scratch)	61.95 ± 2.79	$\mathbf{79.59 \pm 1.83}$
Continual pretraining on OpenWhistle (ours)			
AVES-bio†	AVES-bio checkpoint $\rightarrow$ OpenWhistle	$\mathbf{64.75 \pm 2.64}$	73.87 ± 2.06

Table 1: Performance on whistle-type classification (macro-F1) and whistle-type detection (mAP). Models are grouped by representation and pretraining strategy. All use linear probing with a logistic regression classifier on frozen representations. Scores are reported as percentages. Each entry gives the full-test metric ± the standard deviation over 1,000 bootstrap resamples of the test set. †Continual pretraining at 16 kHz with multi-band encoding; ‡trained natively at 44.1 kHz.